Metaheuristics for University Course Timetabling

STATE-OF-THE-ART REVIEW AND REPRODUCIBLE COMPARISON OF GENETIC ALGORITHM AND SIMULATED ANNEALING

Universities rebuild teaching schedules every semester. The task is routine in appearance but difficult in practice: lectures must be assigned to rooms and periods without conflicts, while preferences and resource limits still matter. We propose a focused review of recent metaheuristic research, followed by a controlled experiment on a public benchmark. The purpose is not to introduce another elaborate hybrid, but to establish what two standard methods can achieve when they are implemented and tested under the same conditions.

RATIONALE AND RESEARCH QUESTIONS

The University Course Timetabling Problem (UCTP) is NP-hard, so large instances are commonly addressed with metaheuristics. Abdipoor et al. [1] report a clear move towards hybrid algorithms, but results remain difficult to compare when papers use different instances, time limits, parameter settings, or reporting standards. That comparability problem gives the project a precise methodological focus.

The study asks two questions. RQ1: Which algorithms, benchmarks, evaluation practices, and unresolved limitations define the current state of the art in university course timetabling? RQ2: When computational budget and evaluation criteria are held constant, how do a Genetic Algorithm (GA) and Simulated Annealing (SA) differ in solution quality, robustness, and running time?

RESEARCH DESIGN

PHASE 1	Literature review. We will search Scopus, Web of Science, IEEE Xplore, ACM Digital Library, and ScienceDirect for peer-reviewed studies published between 2015 and 2026. Earlier foundational papers will be recovered by backward citation searching. Search terms will combine university course timetabling with metaheuristic, genetic algorithm, simulated annealing, local search, hybrid, and benchmark. Duplicate removal, title and abstract screening, full-text assessment, and exclusions will be recorded in a PRISMA-style flow diagram.
PHASE 2	Critical synthesis. Each paper will be coded by problem variant, dataset, constraints, representation, search operators, parameter tuning, stopping rule, baselines, metrics, statistical analysis, code availability, and stated limitations. A quality checklist will distinguish reproducible comparisons from results that cannot be independently verified. The review will organize evidence by method family rather than summarize papers one by one.
PHASE 3	Implementation and benchmark. GA and SA will share the same parser, timetable representation, objective function, initialization procedure, and move/swap neighborhoods. GA will use tournament selection, crossover, mutation, elitism, and repair. SA will use a cooling schedule, the common neighborhoods, and the same repair logic. A constructive greedy solver will serve as a baseline. Keeping these implementations small makes their behaviour easier to interpret and reproduce.

EXPERIMENTAL PROTOCOL

The experiment will use the 21 public instances from Track 3 of the Second International Timetabling Competition (ITC-2007), built from University of Udine data [2]. The official validator separates feasibility from four soft penalties: room capacity, minimum working days, curriculum compactness, and room stability. All methods will receive the same evaluation or time budget. Parameters will be tuned on a separate subset and fixed before final testing; each algorithm will then be run with at least ten independent seeds per instance.

We will report feasibility rate, official penalty, relative gap to published best-known results, time to best solution, convergence, and dispersion across runs. GA and SA will be compared with paired Wilcoxon tests, confidence intervals, and an effect-size measure. Convergence plots and per-instance tables will support the numerical analysis; unsuccessful runs will be retained rather than silently discarded.

EXPECTED CONTRIBUTION, FEASIBILITY, AND LIMITS

The final paper will connect two kinds of evidence: what the literature claims and what a common benchmark shows. Its practical output will be a documented experimental framework with fixed seeds and parameter files. The scope is deliberately narrow enough for the course: one problem formulation, two established algorithms, and one simple baseline. ITC-2007 is public and anonymized, so the project uses no personal data. The discussion will address parameter sensitivity, stochastic variation, machine-dependent timing, and the limits of generalizing from a competition benchmark to a university's live scheduling process.

REFERENCES

[1] S. Abdipoor, R. Yaakob, S. L. Goh, and S. Abdullah, 'Meta-heuristic approaches for the University Course Timetabling Problem,' Intelligent Systems with Applications, vol. 19, 200253, 2023. doi:10.1016/j.iswa.2023.200253.

[2] L. Di Gaspero, B. McCollum, and A. Schaerf, 'The Second International Timetabling Competition (ITC-2007): Curriculum-based Course Timetabling (Track 3),' Technical Report QUB/IEEE/Tech/ITC2007/CurriculumCTT/v1.0, 2007.

[3] T. Muller, 'ITC2007 solver description: a hybrid approach,' Annals of Operations Research, vol. 172, pp. 429-446, 2009. doi:10.1007/s10479-009-0644-y.

[4] R. Lewis and J. Thompson, 'Analysing the effects of solution space connectivity with an effective metaheuristic for the course timetabling problem,' European Journal of Operational Research, vol. 240, no. 3, pp. 637-648, 2015. doi:10.1016/j.ejor.2014.07.041.